THE REAL-ROOTEDNESS PROBABILITY OF A RANDOM CUBIC POLYNOMIAL: THREE CLOSED FORMS

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Abstract. We compute, in closed form, the probability that a random cubic polynomial has three real roots, in three related models. For a *monic* cubic $x^3 + ax^2 + bx + c$ with (a, b, c) independent and uniform on $[-1, 1]$, this probability is $\frac{383}{4860} + \frac{\ln 3}{48}$; on $[0, 1]$ it is exactly $\frac{1}{2880}$. For the *non-monic* cubic ax^3+bx^2+cx+d with all four coefficients independent and uniform on $[-1, 1]$, it is $\frac{641}{2430} - \frac{\ln 3}{24}$. The first two values appear to be new. The third confirms, with a first proof, a closed form that had circulated unproved in an informal online forum thread since June 2026. All three results are verified by independent symbolic and numerical computation; the monic-cubic theorems are additionally verified by a complete, zero-`sorry` machine-checked proof in Lean 4 / Mathlib, and a machine-checked proof of the non-monic theorem is in progress at the time of writing. We record the reduction technique for each case – an elementary band/discriminant-sign argument for the monic case, and a divergence-theorem argument exploiting the homogeneity of the discriminant for the non-monic case – since both appear to generalize to related open problems, which we list.

Contents

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With assistance from Claude (Anthropic). Machinery, proofs, and their verification (symbolic, numerical, and partially machine-checked in Lean 4 / Mathlib) are recorded in the accompanying repository.

1. Introduction

How likely is a "random" polynomial to have all of its roots real? For degree 2 this is classical. For a monic quadratic $x^2 + bx + c$ with (b, c) uniform on $[-1, 1]^2$, real roots occur exactly when $c \le b^2/4$, giving probability 13/24; for the full quadratic ax^2+bx+c the answer involves a logarithm, $\frac{41}{72} + \frac{\ln 2}{12}$ on $[-1, 1]^3$ and $\frac{5}{36} + \frac{\ln 2}{6}$ on $[0, 1]^3$, the latter apparently first recorded (without derivation) by J. D'Aurizio and rederived in [3]. For the analogous *cubic* question the literature is far thinner. The only paper we could locate that claims exact results for random cubics with general coefficient distributions is Li [1], which is effectively unobtainable (two citations in thirty-eight years, neither reproducing a value) and, per the recovered *Mathematical Reviews* notice [2], treats a family of symmetric-interval distributions that does not include the one-sided case of our Theorem 2 at all, and reaches our Theorem 1 only as an unverifiable special case.

This note settles three instances of the cubic question completely.

Theorem 1 (Monic cubic, symmetric cube)**.** *Let (a, b, c) be independent and uniform on $[-1, 1]$. Then*

$$\mathbb{P}\big(x^3 + ax^2 + bx + c \textit{ has three real roots}\big) = \frac{383}{4860} + \frac{\ln 3}{48} = 0.10169434037605886959954086\ldots$$

Theorem 2 (Monic cubic, unit cube)**.** *Let (a, b, c) be independent and uniform on $[0, 1]$. Then*

$$\mathbb{P}\big(x^3 + ax^2 + bx + c \textit{ has three real roots}\big) = \frac{1}{2880}.$$

Theorem 3 (Non-monic cubic)**.** *Let (a, b, c, d) be independent and uniform on $[-1, 1]$. Then*

$$\mathbb{P}\big(ax^3+bx^2+cx+d \textit{ has three real roots}\big) = \frac{641}{2430} - \frac{\ln 3}{24} = 0.21801049620261477108898412335868\ldots$$

Theorems 1 and 2 are, to our knowledge, new: an eight-agent literature sweep together with a targeted search for the exact constants (Section 6) found no prior publication of either value, digit string, or exact expression. Theorem 3 is more subtle. The exact value $\frac{641}{2430} - \frac{\ln 3}{24}$ already existed, unproved, as the terminus of a long, informally written, partly AI-assisted derivation posted to a Russian mathematics forum (`dxdy.ru`) in June 2026. Section 4 gives the first proof of this value, together with five independent lines of numerical and symbolic corroboration.

Motivation and context. The cubic question is the tractable sibling of a harder open problem: for an $n \times n$ matrix with i.i.d. uniform entries, what is the probability that all n eigenvalues are real? For $n = 2$ this is classical (49/72 for entries uniform on $[-1, 1]$, [4]), but already for $n = 3$ no closed form or rigorous nontrivial bound is known; the canonical statement of the question, posted to Mathematics Stack Exchange in 2020, has 153 upvotes and, as of this writing, zero answers [6]. The obstruction is structural: for Gaussian entries, orthogonal invariance gives Edelman's exact $2^{-n(n-1)/4}$ [5], but the uniform cube has no such symmetry to exploit. Moving from matrix-entry space to *coefficient* space – asking instead for the probability that a random *characteristic polynomial* is real-rooted – keeps the same underlying object (the volume of a semialgebraic region cut out by a discriminant) while dropping the ambient dimension from 9 (a 3×3 matrix) to 3–4 (a cubic's coefficients), where it becomes tractable by elementary means. This note is the result of attacking that lower-dimensional sibling.

Machine verification. Every calculus step in the proofs of Theorems 1 and 2 is checked twice over in exact arithmetic (once by hand below, once independently by computer algebra), confirmed by high-precision quadrature and by direct Monte Carlo sampling, and – new for a paper at this level of the literature – verified by a *complete, zero-*`sorry` *machine-checked proof* in Lean 4 against Mathlib, described in Section 5. A machine-checked statement (and partial proof) of Theorem 3 is in progress at the time of writing; we report its status honestly rather than claim completion prematurely.

2. Preliminaries

Throughout, Δ denotes the discriminant of a cubic. For the general cubic ax^3+bx^2+cx+d,

$$\Delta(a,b,c,d)=18abcd-4b^3d+b^2c^2-4ac^3-27a^2d^2, \tag{1}$$

and classically, for $a\neq 0$: $\Delta>0$ iff the cubic has three distinct real roots; $\Delta=0$ iff it has a repeated root (all real); $\Delta<0$ iff it has one real root and a complex-conjugate pair. For the monic case $a=1$ we write $\Delta_3(b,c,d)$ for $\Delta(1,b,c,d)$; abusing notation slightly to match the variable names used in Theorems 1–2 (whose monic cubic is x^3+ax^2+bx+c, i.e. our $(b,c,d)\mapsto(a,b,c)$),

$$\Delta_3(a,b,c)=18abc-4a^3c+a^2b^2-4b^3-27c^2. \tag{2}$$

3. The monic cubic

3.1. **The band reduction.** Write $f(x)=x^3+ax^2+bx+c$ and $g(x)=x^3+ax^2+bx=f(x)-c$. For $b<a^2/3$, put $s=\sqrt{a^2-3b}$; the critical points of f (equivalently of g) are

$$x_-=\frac{-a-s}{3}\ \text{(local max)},\qquad x_+=\frac{-a+s}{3}\ \text{(local min)}.$$

f has three real roots iff $f(x_-)\geq 0\geq f(x_+)$, i.e. iff c lies in the band

$$c\in[c_{\text{lo}},c_{\text{hi}}],\qquad c_{\text{lo}}=-g(x_-),\quad c_{\text{hi}}=-g(x_+),\qquad c_{\text{hi}}-c_{\text{lo}}=\frac{4}{27}s^3.$$

This is not merely an implication but captures the discriminant exactly: as a polynomial in c,

$$\Delta_3(a,b,c)=-27\,(c-c_{\text{lo}})(c-c_{\text{hi}}). \tag{3}$$

(3) is verified by direct expansion (and, independently, by `sympy`). For $b>a^2/3$ the right-hand quadratic in c has no real roots and $\Delta_3<0$ identically (one real root). Hence for any c-window $[w_{\text{lo}},w_{\text{hi}}]$,

$$\text{vol}=\iint_{b<a^2/3}\operatorname{len}\big([c_{\text{lo}},c_{\text{hi}}]\cap[w_{\text{lo}},w_{\text{hi}}]\big)\,db\,da. \tag{4}$$

3.2. **Proof of Theorem 1.**

Lemma 4 (Never-clipped)**.** *On $D=\{(a,b):a\in[-1,1],\ b\in[-1,a^2/3]\}$, $c_{\text{hi}}\leq 1$, with equality only at $(a,b)=(-1,-1)$.*

Proof. Since $\partial g/\partial a$ at fixed x equals x^2 and $g'(x_+)=0$ kills the chain-rule term at the critical point, $\partial c_{\text{hi}}/\partial a=-x_+^2\leq 0$; so c_{hi} is non-increasing in a. As $b\leq a^2/3\leq 1/3$ for $a\in[-1,1]$, the point $(-1,b)$ is also in D for every $(a,b)\in D$, so $c_{\text{hi}}(a,b)\leq c_{\text{hi}}(-1,b)$. On the edge $a=-1$: $\partial c_{\text{hi}}/\partial b=-x_+$ with $x_+=(1+\sqrt{1-3b})/3>0$, so c_{hi} is strictly decreasing in b there, with maximum at $b=-1$: $c_{\text{hi}}(-1,-1)=1$ exactly. $\square$

The map $(a,b,c)\mapsto(-a,b,-c)$ preserves Δ_3 (direct expansion) and the cube, and sends c_{hi} to $-c_{\text{lo}}$; hence $c_{\text{lo}}\geq -1$ on D as well. **The band never exits the window** $[-1,1]$: it touches the boundary only at the two corner points $(\mp 1,-1,\pm 1)$, of measure zero. Therefore, using (4) with no clipping correction,

$$\text{vol}=\int_{-1}^{1}\int_{-1}^{a^2/3}\frac{4}{27}(a^2-3b)^{3/2}\,db\,da=\frac{8}{405}\int_{-1}^{1}(a^2+3)^{5/2}\,da=\frac{3064}{1215}+\frac{8}{3}\operatorname{arsinh}\frac{1}{\sqrt{3}}.$$

Since $\operatorname{arsinh}(1/\sqrt{3})=\ln(1/\sqrt{3}+2/\sqrt{3})=\frac{1}{2}\ln 3$,

$$\mathbb{P}=\text{vol}/8=\frac{383}{4860}+\frac{\ln 3}{48}.$$

Remark 5. The corner-touching geometry is exactly why this is solvable in closed form: the naive "main term plus clipping corrections" program has *all* corrections identically zero, so the trivial upper bound already equals the answer. High-precision quadrature landing on that upper bound to sixteen digits was, in practice, the signal that this was happening.

3.3. **Proof of Theorem 2.** Now the domain is $a \in [0,1]$, $b \in [0, a^2/3]$, window $[0,1]$. Since $b \geq 0$ forces $s \leq a$, both critical points satisfy $x_- \leq 0 \leq$ (the argument of the local max) and $x_+ \leq 0$; and $c_{\mathrm{hi}} \geq 0$ because $g(0) = 0$, $g'(0) = b \geq 0$, and $x_+ \leq 0$ is the local minimum, so $g(x_+) \leq 0$. A short computation shows $c_{\mathrm{hi}} \leq 1/27$ throughout the domain, so the top of the window $[0,1]$ never binds. The bottom does: $\partial c_{\mathrm{lo}}/\partial b = -x_- \geq 0$, and $c_{\mathrm{lo}} = 0$ exactly on the curve $b = a^2/4$ (where $x_- = -a/2$ and $g(-a/2) = 0$); so $c_{\mathrm{lo}} \leq 0$ for $b \leq a^2/4$ and $c_{\mathrm{lo}} \geq 0$ for $a^2/4 \leq b \leq a^2/3$ – a single sign change. Hence the overlap length is c_{hi} on $[0, a^2/4]$ and the full band width on $[a^2/4, a^2/3]$, and, remarkably, all fractional powers cancel:

$$\int_0^{a^2/4} c_{\mathrm{hi}}\, db + \int_{a^2/4}^{a^2/3} \frac{4}{27}(a^2 - 3b)^{3/2}\, db = \frac{a^5}{480}.$$

Integrating over $a \in [0,1]$,

$$\mathbb{P} = \int_0^1 \frac{a^5}{480}\, da = \frac{1}{2880}.$$

4. The non-monic cubic

Dropping the monic restriction – letting the leading coefficient a range uniformly over $[-1,1]$, including near 0 – makes the problem genuinely four-dimensional, and Lemma 4's trick does not generalize directly: it is special to a symmetric window of half-width *exactly* 1 around a *fixed* leading coefficient.

Writing $V(t)$ for the volume of the real-rooted region inside the cube $[-t,t]^3$ in the monic-cubic coefficient space (so $V(1) = 8 \cdot \mathbb{P}$ of Theorem 1), the natural attack conditions on the leading coefficient and rescales:

$$\mathbb{P}(\text{non-monic}) = \int_0^1 \frac{a^3}{8}\, V(1/a)\, da. \tag{5}$$

This requires $V(t)$ for every $t \geq 1$, and (5) turns out to be a dead end: high-precision numerics for $p(a) := a^3 V(1/a)/8$ (computed to 40 digits; e.g. $p(1/2) = 0.1732556347784\ldots$ and $p(1/3) = 0.2277312312100\ldots$) show no recognizable closed form under a blind PSLQ search against $\{1, \ln 2, \ln 3, \ln 5, \sqrt{2}, \sqrt{3}, \pi\}$. $V(1)$ is elementary only because $t = 1$ is exactly where the corner-touching geometry of Lemma 4 applies – not because $V(t)$ is nice in general.

4.1. **The discriminant region is a cone.** The route that works is structural rather than a direct generalization of the band calculus. The discriminant (1) is homogeneous of degree 4: $\Delta(\lambda x) = \lambda^4 \Delta(x)$ for $x = (a,b,c,d)$, so the real-rooted set $R = \{\Delta > 0\}$ is invariant under $x \mapsto \lambda x$ ($\lambda > 0$) – a cone. Apply the divergence theorem to $F = x/4$ (so $\operatorname{div} F = 1$) on $R \cap [-1,1]^4$. Euler's identity gives $x \cdot \nabla\Delta = 4\Delta$, which vanishes identically on the lateral boundary $\{\Delta = 0\}$; so that part of the surface integral is zero, and only the eight faces of the cube contribute:

$$\mathrm{vol}_4(R \cap [-1,1]^4) = \tfrac{1}{4} \oint_{\partial(R \cap [-1,1]^4)} F \cdot n\, dS = \tfrac{1}{2}(S_a + S_b + S_c + S_d),$$

where S_a, S_b, S_c, S_d are the (signed, outward) contributions of the four faces $a = \pm 1, \ldots$ paired by central symmetry $x \mapsto -x$ (which preserves Δ and the cube). The coefficient reversal $(a,b,c,d) \mapsto (d,c,b,a)$ – sending each root x to its reciprocal $1/x$ – also preserves both R and the cube, and identifies $S_a = S_d$, $S_b = S_c$. Hence

$$\mathrm{vol}_4(R \cap [-1,1]^4) = V(1) + S_b, \qquad \mathbb{P}(\text{non-monic}) = \frac{V(1) + S_b}{16}, \tag{6}$$

where $V(1) = S_a = \frac{766}{1215} + \frac{\ln 3}{6}$ (eight times Theorem 1) and

$$S_b = \mathrm{vol}_3\big\{(a, c, d) \in [-1, 1]^3 : ax^3 + x^2 + cx + d \text{ has three real roots}\big\}$$

is the volume on the face where the *second* coefficient is pinned to 1 – a genuinely new, but purely three-dimensional, problem, with neither a $t \to \infty$ tail nor an $a \to 0$ endpoint singularity to contend with.

Remark 6. We verified the reduction step (6) itself against a case with an independently known answer: applied to $q^2 > 4pr$ on $[-1, 1]^3$ (also a cone, the quadratic discriminant being homogeneous), the three faces give $S_p = S_r = 13/6$, $S_q = 5/2 + \ln 2$, and $\frac{1}{3} \cdot 2(S_p + S_q + S_r)$ works out to $\frac{41}{9} + \frac{2}{3} \ln 2$ – exactly eight times the known full-quadratic volume $\frac{41}{72} + \frac{\ln 2}{12}$. The reduction is an identity, not an approximation.

4.2. **Computing S_b.** Substitute $s = \sqrt{1 - 3ac}$ (so $c = (1 - s^2)/(3a)$). The admissible-d band on this face is $[-K_-/u,\ K_+/u]$ with

$$K_+ = (s - 1)^2(2s + 1), \qquad K_- = (s + 1)^2(2s - 1), \qquad u = 27a^2, \qquad K_+ + K_- = 4s^3,$$

on the domain $\{s \in (0, 2),\ a \in [|s^2 - 1|/3, 1]\}$. Integrating over a *first*, at fixed s, is elementary, since a enters only through $u = 27a^2$; two lemmas make it precise.

Lemma 7 (Never clips above)**.** *$d_{\mathrm{hi}} \le 1$ throughout: $\alpha_+ := \sqrt{K_+/27} < a_0 := |s^2 - 1|/3$ reduces to $2s + 1 < 3(s + 1)^2$, true for all s. This is the exact analogue of Lemma 4.*

Lemma 8 (Clips below exactly for $s \in (2/3, 2)$)**.** *$\alpha_- := \sqrt{K_-/27} > a_0$ reduces to $(3s-2)(s-2) < 0$; and $\alpha_- < 1$ reduces to $(s + 1)^2(2s - 1) < 27$, i.e. $s < 2$, with equality precisely at $s = 2$.*

With Lemmas 7–8, $F(s) := \int_{a_0}^{1} L(a, s)/a \, da$ (the a-integral of the band length) is elementary – the term $-K_+/(2K_-) + 2s^3/K_-$ collapses to the constant $1/2$ via $K_+ + K_- = 4s^3$ – giving the piecewise closed form

$$F(s) = \begin{cases} \dfrac{2s^3}{27}\left(\dfrac{1}{a_0^2} - 1\right), & s \le 2/3, \\[2ex] \dfrac{K_+}{54a_0^2} + \dfrac{1}{2} + \ln\dfrac{\alpha_-}{a_0} - \dfrac{2s^3}{27}, & 2/3 < s < 2, \end{cases}$$

continuous at $s = 2/3$ (both branches equal 11264/18225 there). Then

$$S_b = \frac{4}{3}\int_0^2 s\, F(s)\, ds = \frac{1454}{405} - \frac{5}{6}\ln 3 = 2.674613216233365\ldots,$$

and, substituting into (6) with $V(1) = \frac{766}{1215} + \frac{\ln 3}{6}$,

$$\mathbb{P}(\text{non-monic}) = \frac{1}{16}\left(\frac{766}{1215} + \frac{\ln 3}{6} + \frac{1454}{405} - \frac{5}{6}\ln 3\right) = \frac{641}{2430} - \frac{\ln 3}{24}.$$

(Confirmed to hold exactly by direct symbolic simplification in `sympy`.)

4.3. **Independent verification.** Two of the checks below are fully independent of each other and of the derivation above; the rest are internal consistency checks of the computation.

1. **Exact symbolic evaluation** (as above) – `sympy` confirms $\mathbb{P} - (\frac{641}{2430} - \frac{\ln 3}{24}) = 0$ identically.
2. **An independent numerical route, sharing no step with the derivation above**: fixing the leading coefficient a and integrating a out *last* (rather than first, as in (6)) – effectively a direct high-precision evaluation of (5), with the $a \to 0$ endpoint singularity and the migrating clipping breakpoints of the inner integral carefully handled – reproduces $0.21801049620261477102\ldots$, agreeing with the exact value to *nineteen* significant digits (Gauss–Legendre and tanh–sinh quadrature, several degrees each, spread 1.6×10^{-18}).

(3) **Blind PSLQ**: given only the 22-digit numeric value of S_b, a PSLQ integer-relation search (no closed form supplied) recovers $S_b = \frac{1454}{405} - \frac{5}{6}\ln 3$ unprompted.
(4) **Monte Carlo**, 4×10^{10} samples on the raw sign of (1) (using no theory from this paper): $0.218008985 \pm 2.1 \times 10^{-6}$, -0.73σ from the exact value.
(5) Double-precision brute-force quadrature (no special handling of any kind): 0.2180105004, 4.2×10^{-9} from the exact value – at that method's own precision limit.

Remark 9. This integrand carries three numerical hazards worth recording, all concentrated at small leading coefficient a and all invisible to a quadrature routine's own convergence estimate: (i) the conditional probability given the leading coefficient has a *square-root* endpoint singularity as $a \to 0$, $p(a) = p(0) - \frac{2}{3}\sqrt{a} + O(a)$ with $p(0) = \frac{1}{2} + \frac{5}{72} + \frac{\ln 2}{12}$ (the random-quadratic limit, since as $a \to 0$ the third root escapes to $-b/a$), on which a fixed-degree Gauss rule loses roughly seven digits while reporting no warning; (ii) the clipping/kink structure in the inner variable migrates to scale $\sim \sqrt{3a}$ as $a \to 0$, so any fixed panel layout eventually walks past it; and (iii) floating-point arithmetic loses the clipping breakpoints altogether for small a, since the relevant breakpoint cubic has a near-double root whose branches separate by only $O(\sqrt{a})$. All three had to be identified and handled to reach the nineteen-digit confirmation above.

4.4. **Relation to the `dxdy.ru` thread.** The forum thread's stated intermediate volume, $V_0 = \frac{766}{1215} + \frac{\ln 3}{6}$, agrees with $V(1) = 8\cdot$ Theorem 1, and its final claimed value, $\frac{641}{2430} - \frac{\ln 3}{24}$, is confirmed exactly by Theorem 3. Its remaining steps therefore do reach the right answer, though nothing in the present paper validates the specific route the thread took to get there; the derivation above is independent.

5. Formal verification in Lean

Theorems 1 and 2 have been formalized and *completely, machine-checked proved* in Lean 4 against Mathlib (toolchain `v4.33.0`), with **zero** uses of `sorry` and no unexpected axioms. The formal statements are volume/measure statements about the discriminant (2) on the relevant cube, matching (3)–(4) directly; the accompanying repository contains the full source (`NonmonicCubic/Basic.lean`, `Theorem1.lean`, `Theorem2.lean`, `Integrals.lean`).

One presentational difference is worth recording. The informal proof above derives the band via the critical points $x_\pm$ of g. In Lean, it proved cheaper to skip $x_\pm$ entirely and use the single `ring`-provable identity

$$-27\,\Delta_3(a,b,c) = (27c - 9ab + 2a^3)^2 - 4(a^2 - 3b)^3,$$

which yields both branches of (3) at once, and to prove Lemma 4 from the closed forms $27c_{\text{hi}} = (a-s)^2(a+2s)$, $27c_{\text{lo}} = (a+s)^2(a-2s)$ directly, reducing the whole lemma to a single scalar inequality on $a \in [-1,1]$, $s \in [0,2]$ that `nlinarith` discharges automatically with product hints. Both routes are the same mathematics; only the presentation differs, and nothing in the informal proof was found to be incorrect in the process of formalizing it.

A machine-checked statement of Theorem 3 is in progress; at the time of writing, the file `Theorem3 Statement.lean` type-checks and states the theorem precisely, and work continues on a full proof following the elementary (non-divergence-theorem) numerical route of the paper's verification item 2 above, which is expected to be substantially more tractable to formalize than the geometric argument of §4. We will update the accompanying repository as this progresses, and report the final status – complete or partial, with any remaining `sorry`s explicitly enumerated – rather than claim completion here.

6. Novelty and prior art

An eight-agent parallel literature sweep (arXiv, journal databases, Math StackExchange and MathOverflow's own search indices, OEIS, and general web search), followed by a targeted search on the exact constants derived here, found:

- **Theorem 1**: no prior statement of this probability anywhere searchable (OEIS direct digit-sequence query: no match; Math StackExchange/MathOverflow site search for the monic cubic: zero threads; web search for the exact fraction and decimal: empty). Two caveats. First, the *unnormalized* volume $V_0 = \frac{766}{1215} + \frac{\ln 3}{6}$ (i.e. eight times our value) appears, uninterpreted as a probability, as an intermediate step of the `dxdy.ru` thread discussed above – an unrefereed, partly AI-assisted forum derivation of the *non-monic* cubic, dated June 2026, not indexed by any search engine at the time of our search. Second, Li [1] claims an exact expression for the cubic family $x^3 + 3ax^2 + 3bx + 2c$ with (a, b, c) uniform on symmetric intervals $[-h, h] \times [-k, k] \times [-l, l]$; at $h = k = 1/3$, $l = 1/2$ this specializes to exactly our model. The paper itself is paywalled and, per its recovered *Mathematical Reviews* notice [2], no numerical or symbolic value from it has ever been reproduced by either of its two citations; whether it correctly reduces to Theorem 1 cannot be checked without the original text.
- **Theorem 2**: likely new without caveat – Li's furnished special cases are all symmetric intervals, and the value does not appear in the `dxdy.ru` thread or anywhere else searched.
- **Theorem 3**: the *value* is not new (see above); the *proof* and its independent verification are.

We regard the Li [1] question as the one genuinely open item in this account.

7. Open problems

The reduction techniques used here appear to generalize.

(1) **Monic quartic.** The quartic discriminant is homogeneous of degree 6, so the divergence-theorem argument of §4 should apply, reducing the 4-dimensional all-real-roots volume to a sum over $2 \times 4 = 8$ three-dimensional face volumes (with the further reduction each of those face volumes admits still to be worked out); our current Monte Carlo estimate for the monic case is $0.0054749 \pm 9.0 \times 10^{-6}$.

(2) **Gaussian coefficients.** For the monic cubic with (a, b, c) i.i.d. standard normal, there is no cube to bound the band, so Lemma 4 does not apply in any form; the natural object is $\mathbb{E}[\Phi(c_{\mathrm{hi}}) - \Phi(c_{\mathrm{lo}})]$, a smooth two-dimensional integral with no case splits, but a priori no reason to expect an elementary closed form. Current Monte Carlo estimate: $0.169962 \pm 4.2 \times 10^{-5}$.

(3) **General degree n.** The probability of no real roots for a degree-n Kac polynomial decays as $n^{-3/4}$ exactly (Poplavskyi–Schehr [7]); the corresponding rate constant for *all* n roots real, under speed-n^2 large-deviation scaling, is not known in closed form for uniform (or Kac-Gaussian) coefficients.

(4) **The parent matrix problem.** For an $n \times n$ matrix with i.i.d. uniform entries, the all-real-eigenvalue probability remains open for $n \geq 3$; the striking empirical near-coincidence $\mathbb{P}_n(\text{uniform}) \approx 2^{-(n-1)(n-2)/4}$ (Edelman's Gaussian probability at rank $n-1$), first observed numerically on Math StackExchange, has no known explanation.

Appendix A. Numerical verification summary

	Theorem 1	Theorem 2	Theorem 3
Symbolic (`sympy`, exact)	PASS	PASS	PASS ($\mathbb{P}-$ cand. $=0$)
Independent numerical route	$\sim 10^{-16}$	$\sim 10^{-13}$	19 sig. digits (7.0×10^{-20})
Blind PSLQ / `identify`	—	—	recovers exact form
Monte Carlo	2×10^{8}: $+1.3\sigma$	2×10^{8}: -0.5σ	4×10^{10}: -0.73σ
Lean 4 / Mathlib	**0** `sorry`	**0** `sorry`	in progress

Data and code availability

All source code (Python: `sympy`/`mpmath`/`numpy`/`scipy`; Lean 4: Mathlib `v4.33.0`), raw numerical results, the literature-sweep records underlying §6, and an interactive visual explainer of the monic-cubic case are available in the accompanying repository.

References

[1] Hung C. Li, *The exact probability that the roots of quadratic, cubic, and quartic equations are all real if the equation coefficients are random*, Comm. Statist. Theory Methods **17** (1988), no. 2, 395–409.
[2] G. Samal, review of [1], Mathematical Reviews MR89j:60069 (1989).
[3] S. Haldar and S. Chakraborty, *On the Probability of Real Roots in a Quadratic Equation with Coefficients as i.i.d. $U(-\theta,\theta)$ Variates*, J. Appl. Math. Comput. **5** (2021), no. 1, 48–55.
[4] A. J. Hetzel, J. S. Liew, and K. E. Morrison, *The Probability that a Matrix of Integers Is Diagonalizable*, Amer. Math. Monthly **114** (2007), 491–499.
[5] A. Edelman, *The Probability that a Random Real Gaussian Matrix has k Real Eigenvalues, Related Distributions, and the Circular Law*, J. Multivariate Anal. **60** (1997), no. 2, 203–232.
[6] *Probability for an $n\times n$ matrix to have only real eigenvalues*, Mathematics Stack Exchange, question 3770846, asked 2020-07-27.
[7] M. Poplavskyi and G. Schehr, *Exact Persistence Exponent for the 2d-Diffusion Equation and Related Kac Polynomials*, Phys. Rev. Lett. **121** (2018), 150601.